```
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#PROJECT TITLE:Smartphone Usage and Addiction Analysis
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```
  import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

#We imported all necessary libraries for data analysis, visualization, and machine learning.
```

```
df = pd.read_csv("Smartphone_Usage_And_Addiction_Analysis_7500_Rows.csv")

df.head()
#The first 5 rows of the dataset are displayed to understand its structure.
```

	transaction_id	user_id	age	gender	daily_screen_time_hours	social_media_hours	gaming_hours	work_study_hours	sleep_hour
0	TXN00001	U00001	21	Male	3.23	2.01	0.89	4.55	7.5
1	TXN00002	U00002	24	Other	5.09	3.81	2.24	4.44	7.6
2	TXN00003	U00003	31	Other	6.06	1.36	3.83	2.35	4.92
3	TXN00004	U00004	32	Other	7.83	5.85	1.51	3.54	8.2
4	TXN00005	U00005	25	Male	9.96	5.92	3.42	5.27	6.2

```
df.shape
df.dtypes
#The dataset contains 7500 rows and multiple columns with both numeric and categorical data.
```

	0
transaction_id	object
user_id	object
age	int64
gender	object
daily_screen_time_hours	float64
social_media_hours	float64
gaming_hours	float64
work_study_hours	float64
sleep_hours	float64
notifications_per_day	int64
app_opens_per_day	int64
weekend_screen_time	float64
stress_level	object
academic_work_impact	object
addiction_level	object
addicted_label	int64

dtype: object

```
df.isnull().sum()
# Fill numeric missing values with mean
for col in df.select_dtypes(include=np.number).columns:
    df[col].fillna(df[col].mean(), inplace=True)

# Fill categorical with mode
for col in df.select_dtypes(include='object').columns:
    df[col].fillna(df[col].mode()[0], inplace=True)
    df.drop_duplicates(inplace=True)
    #Missing values were handled using mean (numeric) and mode (categorical). Duplicate rows were removed to ensure clean data.
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_4315/3036012567.py in <cell line: 0>()
----> 1 df.isnull().sum()
      2 # Fill numeric missing values with mean
      3 for col in df.select_dtypes(include=np.number).columns:
      4     df[col].fillna(df[col].mean(), inplace=True)
      5

NameError: name 'df' is not defined
```

Next steps: Explain error

```
col1 = df.select_dtypes(include=np.number).columns[0]
col2 = df.select_dtypes(include=np.number).columns[1]

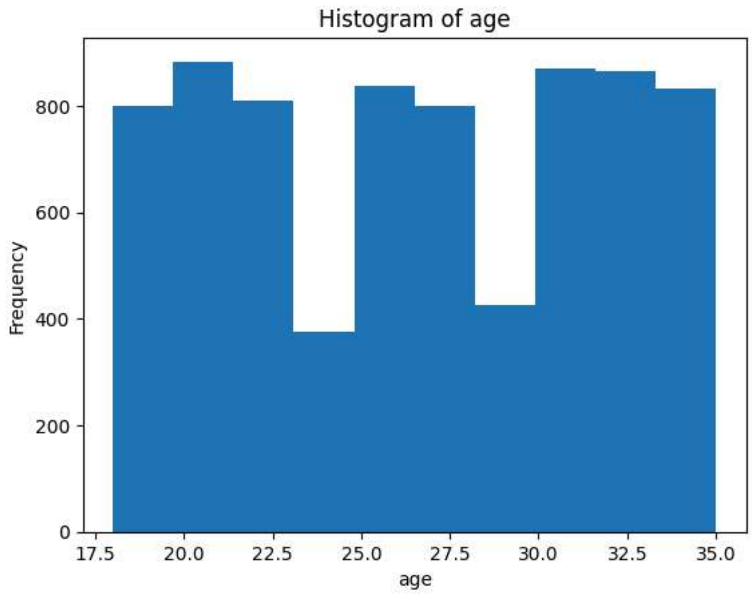
def stats(col):
    return {
        "Mean": df[col].mean(),
        "Median": df[col].median(),
        "Mode": df[col].mode()[0],
        "Std Dev": df[col].std(),
        "Variance": df[col].var(),
        "Range": df[col].max() - df[col].min(),
        "Mid-Range": (df[col].max() + df[col].min()) / 2
    }

stats(col1), stats(col2)

#Statistical measures were calculated to understand distribution, spread, and variability of key numeric columns.
```

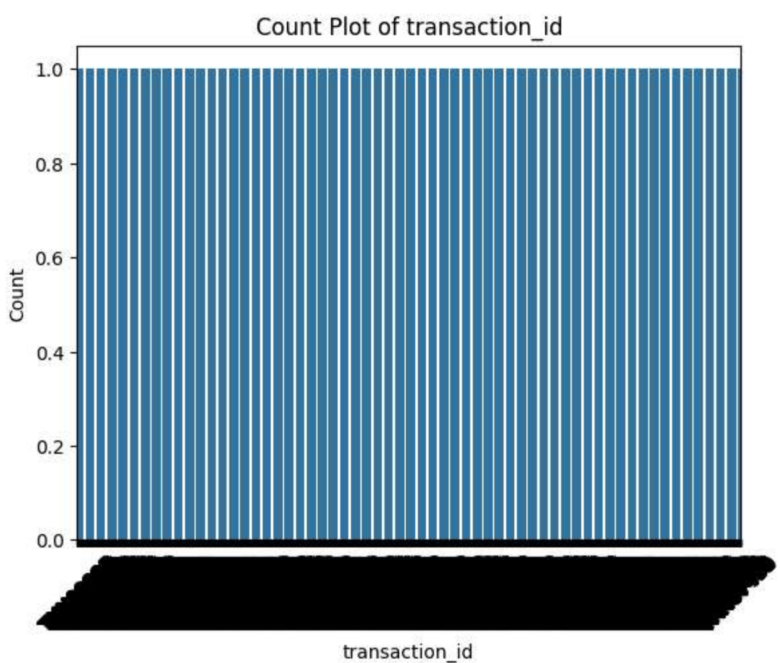
```
({'Mean': np.float64(26.5688),
  'Median': 27.0,
  'Mode': np.int64(31),
  'Std Dev': 5.19710828743003,
  'Variance': 27.009934551273904,
  'Range': 17,
  'Mid-Range': 26.5},
 {'Mean': np.float64(7.499911999999999),
  'Median': 7.525,
  'Mode': np.float64(8.13),
  'Std Dev': 2.609187505753729,
  'Variance': 6.807859440181365,
  'Range': 9.0,
  'Mid-Range': 7.5})
```

```
plt.figure()
plt.hist(df[col1])
plt.title("Histogram of " + col1)
plt.xlabel(col1)
plt.ylabel("Frequency")
plt.show()
#The histogram shows how values of this variable are distributed across users.
```

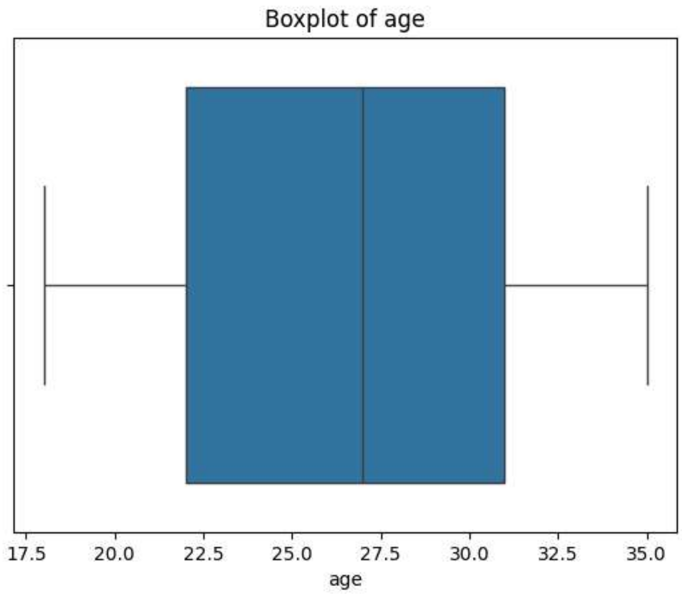



```
cat_col = df.select_dtypes(include='object').columns[0]

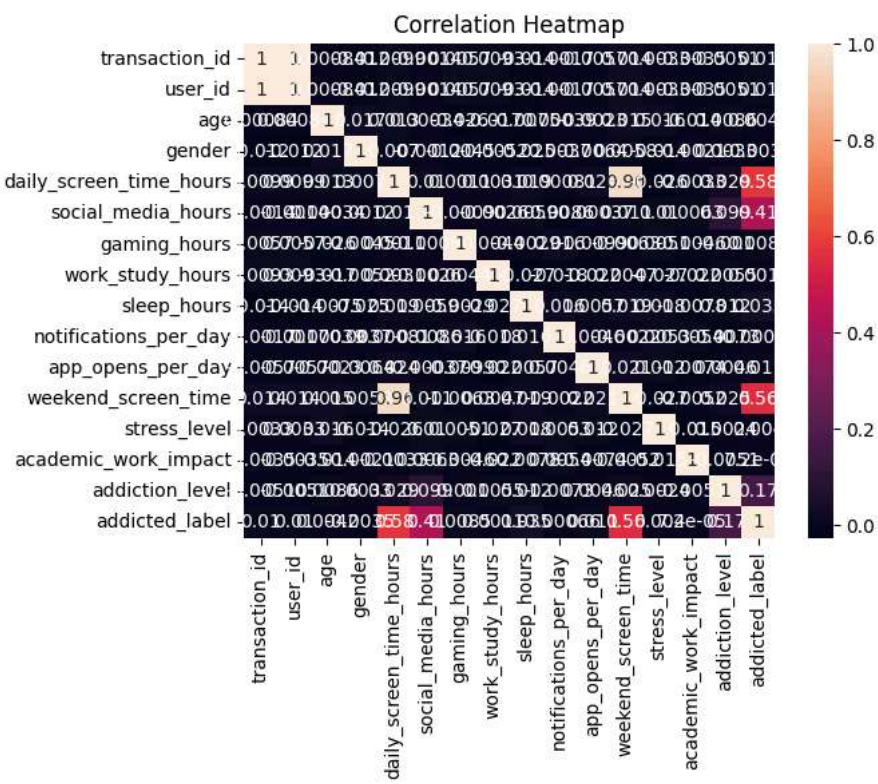
plt.figure()
sns.countplot(x=df[cat_col])
plt.title("Count Plot of " + cat_col)
plt.xlabel(cat_col)
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.show()
#This chart shows the distribution of categories in the dataset.
```




```
plt.figure()
sns.boxplot(x=df[col1])
plt.title("Boxplot of " + col1)
plt.xlabel(col1)
plt.show()
#The boxplot highlights the spread of data and possible outliers.
```




```
plt.figure()
sns.heatmap(df.corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()
#The heatmap shows relationships between numeric variables.
```




```
le = LabelEncoder()

for col in df.select_dtypes(include='object').columns:
    df[col] = le.fit_transform(df[col])
    X = df.drop(df.columns[-1], axis=1)
y = df[df.columns[-1]]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

```
accuracy_score(y_test, y_pred)
#The model predicts the target variable based on usage patterns. The accuracy shows how well the model performs on unseen data.
## Insights

#1. The histogram shows that most users fall within a specific usage range, indicating common usage behavior.

#2. The boxplot reveals some extreme outliers, suggesting a small group of highly addicted users.

#3. The correlation heatmap shows strong relationships between usage time and addiction level.
## Recommendations

#1. Users with extremely high usage should be monitored and encouraged to reduce screen time.

#2. Awareness programs can help users understand the impact of excessive smartphone use.
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge (sta
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
0.872
```